Experiment No : 10

Date:

Title : SRS for Student Management System

AIM:

The objective of this experiment is to develop a Software Requirements Specification (SRS) document for a Student Management System. The aim is to clearly define both functional and non-functional requirements to ensure the system operates efficiently, securely, and reliably.

PROBLEM DESCRIPTION:

Traditional student management processes rely heavily on manual record keeping for attendance, marks, and student details. This leads to errors, redundancy, and difficulty in retrieving data. To overcome these issues, a computerized Student Management System is required.

The system should allow storing student information, managing attendance, recording marks, and generating reports. It should improve accuracy, reduce paperwork, and provide easy access to academic data for students, faculty, and administrators.

CONDITIONS:

The system should include secure login functionality to ensure authorized access. It must support multiple user roles such as admin, faculty, and students.

The application should be web-based and accessible through standard browsers. A database system must securely store student records, attendance, and marks. The system should also handle multiple users simultaneously without performance degradation.

METHODOLOGY:

The development follows standard software engineering practices. Requirements are gathered, analyzed, and documented in SRS format.

The system is divided into modules such as:

- Student Management
- Attendance Management
- Marks Management
- Faculty Management
- Report Generation

Functional requirements define system behavior, while non-functional requirements focus on performance, security, and usability.

PLANNING AND ESTIMATION

The project is divided into phases:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment

Each module is assigned time and resources. Proper estimation ensures timely completion and efficient resource utilization.\

SCHEDULING:

Development is carried out step-by-step:

1. User authentication module
2. Student data management
3. Attendance tracking
4. Marks processing
5. Report generation

The system is designed with focus on:

1. Fast performance
2. Data security
3. Scalability
4. User-friendly interface.

FEASIBILITY ANALYSIS:

Technical Feasibility:
The system is feasible using technologies like Java/Python and MySQL database.

Economic Feasibility:
Development and maintenance costs are low and affordable.

Operational Feasibility:
The system is easy to use and improves efficiency for faculty and students.

RESULT:

The SRS document for the Student Management System has been successfully prepared. It clearly defines system functionalities, requirements, and constraints necessary for development.

CONCLUSION:

The Student Management System provides an effective solution for managing academic activities. The SRS document acts as a foundation for development, reducing errors and ensuring system quality, reliability, and maintainability.